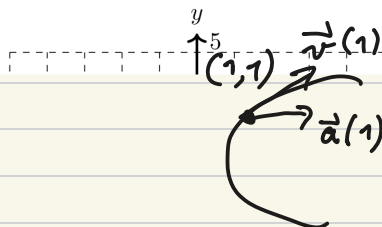


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10. For each of the following paths, find the equations of the velocity $\vec{v}(t) = \vec{x}'(t)$ and of the acceleration $\vec{a}(t) = \vec{v}'(t) = \vec{x}''(t)$. Sketch the path and on the same graph include the velocity vector and acceleration vector at the indicated time t_0 .

a) $\vec{x}(t) = (t^2, t)$ for $-3 \leq t \leq 3$ and $t_0 = 1$



$$\vec{v}(t) = \vec{x}'(t) = (2t, 1)$$

$$\vec{a}(t) = \vec{v}'(t) = \vec{x}''(t) = (2, 0)$$

$$\vec{x}(1) = (1, 1) \quad \vec{v}(1) = (2, 1) \quad \vec{a}(1) = (2, 0)$$

11. Calculate the length of each of the following paths:

a) $\vec{x}(t) = (2t + 1, 7 - 3t)$, $-1 \leq t \leq 2$

$$a = -1, b = 2$$

$$\int_a^b \|\vec{v}(t)\| dt$$

$$\text{length} = s = \int_{-1}^2 \sqrt{13} dt$$

$$\vec{v}(t) = \vec{x}'(t) = (2, -3)$$

$$\|\vec{v}(t)\| = \sqrt{4+9} = \sqrt{13}$$

$$= \sqrt{13} t \Big|_{-1}^2 = 2\sqrt{13} - (-1)\sqrt{13} = 3\sqrt{13}$$

5. A ladybug who is very sensitive to temperature is crawling on graph paper. It is at the point $(3, 7)$ and notices that if it moves in the $\hat{i}$ -direction, the temperature increases at a rate of 3 deg/cm. If it moves in the $\hat{j}$ -direction, it notices the temperature decreases at a rate of 2 deg/cm. In what direction should the ladybug move if:

a) it wants to warm up most rapidly? direction of $(3, -2)$ or $(3, -2)$ or $(\frac{3}{\sqrt{13}}, \frac{-2}{\sqrt{13}})$

b) it wants to cool off most rapidly? direction of $(-3, 2)$ or $(-3, 2)$ or $(\frac{-3}{\sqrt{13}}, \frac{2}{\sqrt{13}})$

c) it desires its temperature not to change? $(2, 3)$ and $(-2, -3)$

$$\vec{\nabla} T(3, 7) = \left(\frac{\partial T}{\partial x} \Big|_{(3, 7)}, \frac{\partial T}{\partial y} \Big|_{(3, 7)} \right) = (3, -2)$$

Recall tangent (line, plane, hyperplane) explicit vs. implicit

e.g. **Explicit**: the curve (or surface, or hypersurface) is the graph of a function



Curve $y = f(x) = \sqrt{1-x^2}$ find eq. of tangent line at $x = \frac{1}{2}$
 $\frac{dy}{dx} = \frac{-x}{\sqrt{1-x^2}}$ $m = \frac{-1/2}{\sqrt{3}/2} = -\frac{1}{\sqrt{3}}$
 $y = f(\frac{1}{2}) + m(x - \frac{1}{2})$
 $y = \frac{\sqrt{3}}{2} + (-1/\sqrt{3})(x - 1/2)$

Implicit: the curve (or surface, or hypersurface) consists of all the points that satisfy an equation

Curve defined by $x^2 + y^2 = 1$ find eq. of tangent line at $(\frac{1}{2}, \frac{\sqrt{3}}{2})$
 say $f(x,y) = x^2 + y^2$ the given curve is the level curve $f(x,y) = 1$
 eq of tangent $0 = \nabla f|_{(\frac{1}{2}, \frac{\sqrt{3}}{2})} \cdot ((x,y) - (\frac{1}{2}, \frac{\sqrt{3}}{2})) = (2 \cdot \frac{1}{2}, 2 \cdot \frac{\sqrt{3}}{2}) \cdot (x - \frac{1}{2}, y - \frac{\sqrt{3}}{2})$

e.g. **Explicit** $z = g(x,y) = \sqrt{1-x^2-y^2}$ find eq. of tangent plane at $(x,y) = (\frac{1}{2}, \frac{1}{2})$
 eq. of tangent plane $z = g(\frac{1}{2}, \frac{1}{2}) + \nabla g|_{(\frac{1}{2}, \frac{1}{2})} \cdot ((x,y) - (\frac{1}{2}, \frac{1}{2}))$
 $\nabla g = (\frac{-x}{\sqrt{1-x^2-y^2}}, \frac{-y}{\sqrt{1-x^2-y^2}})$ $\nabla g|_{(\frac{1}{2}, \frac{1}{2})} = (\frac{-1/2}{\sqrt{1/2}}, \frac{-1/2}{\sqrt{1/2}}) = \frac{1}{\sqrt{2}}(-1, -1)$
 $z = \frac{1}{\sqrt{2}} + (\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}) \cdot (x - \frac{1}{2}, y - \frac{1}{2}) = \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}(x+y) + \frac{1}{\sqrt{2}}$

Implicit $x^2 + y^2 + z^2 = 1$ find eq. of tangent plane at $(\frac{1}{2}, \frac{1}{2}, \frac{1}{\sqrt{2}})$
 $f(x,y,z) = x^2 + y^2 + z^2$ the points satisfying $x^2 + y^2 + z^2 = 1$ are exactly the points in the level $f(x,y,z) = 1$
 $x^2 + y^2 = 1 - z^2 \leadsto x^2 + y^2 + z^2 - 1 = 0$ $h(x,y,z) = x^2 + y^2 + z^2 - 1$
 level set $h(x,y,z) = 0$

eq. of tang. plane: $0 = \nabla h|_{(\frac{1}{2}, \frac{1}{2}, \frac{1}{\sqrt{2}})} \cdot (x - \frac{1}{2}, y - \frac{1}{2}, z - \frac{1}{\sqrt{2}})$
 $\nabla h = (2x, 2y, 2z)$ $= 0 = (1, 1, \sqrt{2}) \cdot (x - \frac{1}{2}, y - \frac{1}{2}, z - \frac{1}{\sqrt{2}}) = x - \frac{1}{2} + y - \frac{1}{2} + \sqrt{2}z - 1$

Total differential

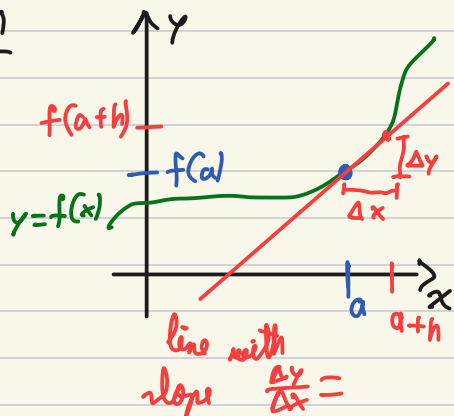
Let's go back to $f(x): \mathbb{R} \rightarrow \mathbb{R}$ as in Calc I.

$$f'(a) = \frac{df}{dx}(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

df we take h very small
(i.e. Δx very small)

$$\Delta y \approx f'(a) \cdot \Delta x$$

↳ i.e. best linear
approximation to $y=f(x)$



when Δx is infinitesimally small we represent this as $dy = f'(a) dx$

def: Suppose $f: X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ is differentiable at $\vec{a}$ in X , then we define the total differential of f by

$$df = \vec{\nabla} f|_{\vec{a}} \cdot (dx_1, dx_2, \dots, dx_n)$$

$$df = \frac{\partial f}{\partial x_1} dx_1 + \frac{\partial f}{\partial x_2} dx_2 + \dots + \frac{\partial f}{\partial x_n} dx_n$$

eg. $f(x, y) = x^3 + 2xy - y^2$

$$\vec{\nabla} f = (3x^2 + 2y, 2x - 2y)$$

$$df = (3x^2 + 2y)dx + (2x - 2y)dy$$